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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

Second Semester M.Tech (CST) (CBCS) Degree Examinations

May – June 2025

22CST232 – DESIGN OF CONCRETE BRIDGES

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

2) Use of code of practice **IRC 6, IRC 21, IS 1343, IS 456** and Pigeaud's chart are permitted.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Write the factors to be considered in selection of site for a bridge	10	L*2	1	1,2
	b) Evaluate the loads acting on top slab, bottom slab and side walls for a box culvert of dimensions 4m x 4m, Subjected to a dead load of 14kN/m ² . Take live load of IRC class AA tracked vehicle, assume unit weight of soil as 18 kN/m ³ , Angle of repose 30°, Road width is 7.5m, with 1m wide footpath on both sides.	10	L3	2	2
2.	a) Draw the neat sketch of a typical bridge and name the components	10	L2	1	1,2
	b) Evaluate fixed end moments for bottom slab and top slab for a box culvert of dimensions 4m x 4m, design loading on top slab is 57.4 kN/m ² , on bottom slab it is 87.4 kN/m ² , on side walls trapezoidal load varies from 15.5 kN/m ² at the top and 42.2 kN/m ² at the bottom.	10	L3	2	2
3.	a) Write an explanatory note on linear water way, afflux and their significance in design of bridges.	10	L2	1	1,2
	b) Explain why box culverts are economical and the critical load cases for design of box culvert.	10	L2	2	2
Unit – II					
4.	a) Write a short note on Pigeaud's, curves, discuss their utilities and limitations	10	L2	3	1,2
	b) A T-beam girder bridge has an effective span of 16m. Clear roadway between kerbs is 7.5m. Kerbs are 600mm wide. Three identical longitudinal girders are at 2.6m c/c have been provided. loading is IRC class A. Determine reaction factor by Courbon's method.	10	L3	4	1,2
5.	a) Obtain the values of long span and short span bending moments for an interior panel of a T beam bridge having following details Dimensions of the panel = 3m x 3.5m Loading IRC class A Loading pattern : 2 wheels (each of 57 kN) adjusted symmetrically with respect to center of the panel.	10	L3	3	1,2
	b) Obtain the expression for courbon's reaction factor and discuss its limitations.	10	L2	4	1,2
6.	a) Explain different methods available for design of interior panel of slab and mention the limitations of Pigeaud's method.	10	L2	3	1,2
	b) Obtain the courbon's reaction factor and maximum bending moment due to live load in case of a T-beam bridge having the following details Road way 7.5m with 1m footpath on either side Live load IRC class AA tracked vehicle Number of girders are 3 with c/c spacing of 3m Span of the bridge = 14m	10	L3	4	1,2

P.T.O.

Unit – III

7. Explain the design steps for a post tensioned PSC girder. 20 L3 5 1,2
8. Design a PSC bridge girder with following details
Grade of concrete is M50
Span is 25m
Cube strength of concrete at transfer is 40N/mm^2
Loss ratio η is 0.85
Class Type 1 structure
Characteristic strength of high tensile wires is $f_{yp} = 1500\text{N/mm}^2$
DL bending moment $M_g = 3008\text{ kN-m}$
LL bending moment $M_L = 1706\text{ kN-m}$
DL shear force $V_g = 600\text{kN}$
LL shear force $V_L = 200\text{kN}$
Calculate the pre-stressing force and area of steel, 20 L3 5 1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
